

Artificial Intelligence

Warehouse Storage Optimization Using Genetic Algorithm

Assignment-2

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1. Chromosome Representation

Each chromosome is a list of 20 integers. Each index represents a product (index 0 = P1, index 19 = P20) and each value represents the zone assigned to that product (1 to 8).

Example:

[3, 4, 5, 1, 3, 4, 5, 6, 3, 5, 4, 1, 3, 5, 6, 5, 4, 3, 7, 5]

Integers were used instead of strings like 'Z1' because they are simpler and can be used directly as dictionary keys.

2. Fitness Function

2.1 Overview

The fitness function calculates the total penalty score for a chromosome. Lower score means better solution. The goal is to minimize it to 0.

2.2 Penalty Table

Constraint	Violation	Penalty
5.1	Zone weight capacity exceeded	+2 per kg over limit
5.2	Fragile item placed outside Z3	+8 per product
5.3	Hazardous item placed outside Z5	+10 per product
5.4	Temp-sensitive item placed outside Z4 or Z8	+9 per product
5.5	High-demand item not placed in Z6	+5 per product
5.6	Heavy item (>40 kg) not placed in Z1	+4 per product
5.7	Compatible items stored in different zones	+3 per separated pair
5.8	Food and Chemical items sharing a zone	+15 per pair
5.9	Non-eligible product placed in Z8	+12 per product

2.3 How Penalties Are Calculated

Pre-computed sets are used for fast lookups (fragile, hazardous, temp-controlled, etc.).

5.1 - Zone Weight Capacity

Total weight per zone is calculated. If it exceeds capacity, penalty = excess kg x 2.

5.2 - Fragile Product Protection

If a fragile product is not in Z3, penalty +8 per product.

5.3 - Hazardous Material Isolation

If a hazardous product is not in Z5, penalty +10 per product.

5.4 - Temperature-Controlled Storage

If a temp-controlled product is not in Z4 or Z8, penalty +9 per product.

5.5 - Fast-Moving Product Accessibility

If a high-demand product is not in Z6, penalty +5 per product. Products with multiple constraints get penalized for each violation independently.

5.6 - Heavy Item Placement

If a product weighing more than 40 kg is not in Z1, penalty +4 per product.

5.7 - Product Compatibility

Every pair of same-category products is checked. If any pair is in different zones, penalty +3 per pair.

5.8 - Food and Chemical Incompatibility

For each zone, $\text{food_count} \times \text{chem_count}$ gives the number of food-chemical pairs sharing that zone. Penalty = pairs $\times$ 15.

5.9 - Loading Dock Restriction

If a non-eligible product is placed in Z8, penalty +12 per product. Only Frozen Meat, Ice Cream, Yogurt Cartons, and Frozen Fish are eligible for Z8.

2.4 Why Fitness = 0 May Not Be Achievable

There are conflicts between constraints that make fitness = 0 impossible:

- High-demand products total 238 kg but Z6 capacity is only 60 kg — weight violation is unavoidable

- Fragile products total 87 kg but Z3 capacity is only 80 kg — another unavoidable weight violation

So the GA minimizes the penalty as much as possible instead of reaching zero.

3. Selection Strategy

3.1 Available Strategies

Roulette Wheel Selection

Each chromosome gets a probability proportional to its inverted fitness. Better chromosomes have higher chance of being selected. Like spinning a weighted wheel.

Tournament Selection

For this, k random chromosomes are picked and the one with lowest fitness wins. This is repeated to select each parent.

Rank Selection

Chromosomes are sorted and assigned ranks. Selection probability is based on rank instead of raw fitness values.

3.2 Chosen Strategy: Tournament Selection

Tournament Selection was used with tournament size $k = 3$.

3.3 Justification

- Fitness can reach 0 which causes division by zero in Roulette Wheel
- Tournament size k is easy to tune — higher k = more pressure toward best chromosomes
- Works well with complex multi-constraint problems

3.4 Tradeoffs

- Roulette Wheel: more mathematically elegant but breaks with zero fitness values
- Rank Selection: more stable but loses information about how different the fitness values actually are

4. Crossover Method

4.1 Available Methods

Single-Point Crossover

One random cut point is chosen. Child1 takes genes before the cut from Parent1 and after the cut from Parent2. Child2 is the reverse.

Two-Point Crossover

Two random cut points are chosen. Child1 takes the outside segments from Parent1 and the middle segment from Parent2. Child2 is the reverse.

4.2 Chosen Method: Two-Point Crossover

Two-Point Crossover was used with pt1 from the first half and pt2 from the second half of the chromosome.

4.3 Justification

- Chromosome is 20 genes long — single-point does not mix genes enough
- Two cut points give better mixing across the whole chromosome
- Less destructive than uniform crossover which could break good partial solutions

4.4 How It Works In The Code

pt1 is chosen from the first half, pt2 from the second half. This guarantees a meaningful middle segment is always swapped.

```
child1 = parent1[:pt1] + parent2[pt1:pt2] + parent1[pt2:]  
child2 = parent2[:pt1] + parent1[pt1:pt2] + parent2[pt2:]
```

5. Mutation Strategy

5.1 Method Used

Random resetting mutation was used. Each gene has a small probability of being replaced with a random zone number (1-8).

5.2 Mutation Rate

Base mutation rate is 5% per gene with an adaptive strategy:

- If no improvement for 50 generations, mutation rate increases by 5%
- Maximum mutation rate is capped at 40%
- When improvement is found, mutation rate resets to 5%

This helps the GA escape local optima when it gets stuck.

6. Population Size and Generation Count

6.1 Population Size

Population size: 200

- Larger population gives more diversity and better exploration
- Assignment suggested 50-200, so 200 was chosen for maximum coverage

6.2 Generation Count

Generation count: 1000

- Enough generations for the GA to converge
- Early termination is also used if fitness reaches 0

6.3 Other Parameters

Parameter	Value	Reason
Tournament Size (k)	3	Balances selection pressure and diversity
Children per Generation	40 (20 pairs)	More new candidates each generation = faster convergence
Elitism	Enabled	Best chromosome always preserved
Adaptive Mutation	Enabled	Increases mutation rate when stuck

7. Results

7.1 Best Fitness Achieved

The best fitness score achieved is 78.

7.2 Optimal Storage Plan

Zone	Zone Name	Products Assigned
Z1	Heavy Item Floor Storage	Rice Bags, Flour Bags
Z2	Standard Rack Storage	(empty)
Z3	Fragile Item Shelf	Glass Bottles, Ceramic Plates, Olive Oil Bottles, Wine Bottles, Bubble Wrap Rolls
Z4	Temperature Controlled Storage	(empty)
Z5	Hazardous Material Storage	Cleaning Acid, Detergent, Industrial Bleach, Paint Cans, Hand Sanitizer
Z6	Fast-Moving Product Area	Chips Carton, Biscuit Boxes, Wheat Sacks
Z7	Bulk Dry Storage	Motor Oil
Z8	Refrigerated Loading Dock	Frozen Meat, Ice Cream, Yogurt Cartons, Frozen Fish

7.3 Remaining Violations

Some violations remain because of the conflicts explained in Section 2.4:

- Constraint 5.5: Temp-controlled products (Frozen Meat, Ice Cream, etc.) must go to Z8 due to constraint 5.4, so they cannot also be in Z6. Rice Bags and Flour Bags must go to Z1 due to constraint 5.6. These conflicts make fully satisfying 5.5 impossible.
- Constraint 5.7: Motor Oil is in Z7 instead of Z5 because Z5 is full. This causes separated-pair penalties with other Chemical products.
- Constraint 5.1: Z6 and Z3 weight limits cannot be avoided given the number of products that must go there.

8. Bonus Task - Excel Report

The `export_to_excel()` function saves the final storage plan to an Excel file (`optimized_warehouse.xlsx`) using `pandas` and `openpyxl`. The file has 3 sheets:

- **Product Allocation:** shows every product with its assigned zone and all its properties. "Yes" values are highlighted in green.
- **Zone Summary:** shows each zone's capacity, used weight, remaining space, and all products in it. Rows turn red if a zone is over capacity.
- **Summary:** shows the best fitness score and overall stats.